Dense Retrieval vs. Graph RAG: Evaluating Factuality and Hallucination in Complex Medical Question Answering

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ABSTRACT

Retrieval-Augmented Generation (RAG) is increasingly deployed in clinical decision support. While dense vector retrieval models (such as Contriever and DPR) are standard, their capacity for multi-hop relational inference remains contested. In this empirical study across 10,000 PubMed QA pairs, we demonstrate that Graph-augmented retrieval reduces medical factual hallucination by 34.2% compared to standard dense vector retrieval. However, structured knowledge graph traversal incurs a substantial 4.1x latency penalty (mean 420ms vs 102ms), challenging real-time bedside deployment. We provide statistical bounds and analyze failure modes where vector embeddings lose relational context across disparate medical records.

1. Introduction

Large language models (LLMs) continue to suffer from catastrophic factual hallucinations in clinical medicine. Standard dense retrieval encodes documents into continuous latent vectors, measuring semantic proximity. However, biomedical relationships often require explicit symbolic traversal (e.g., drug-interaction pathways).

2. Methodology & Experimental Setup

We constructed an enterprise biomedical knowledge graph using Neo4j over 2.4 million PubMed articles. We compared Dense Passage Retrieval (DPR) with our hybrid GraphRAG pipeline using GPT-4 and Med-PaLM-2 as generation backbones. Evaluation metrics include Factual Precision, Hallucination Rate (HR), and Inference Latency across 10,000 queries.

3. Key Empirical Findings & Claims

- **Finding 1: Graph-augmented retrieval reduces medical factual hallucination by 34.2% compared to standard dense vector retrieval.**
- **Finding 2: Vector embeddings alone struggle with multi-hop diagnostic reasoning across disparate patient history nodes, showing a 42% accuracy collapse on 3-hop queries.**
- **Finding 3: Knowledge graph retrieval introduces a 4.1x latency penalty, making real-time clinical deployment cost-prohibitive on commodity hardware.**

4. Limitations & Research Gaps

Our study is strictly restricted to static English medical literature. We identify a critical gap: existing graph RAG frameworks cannot ingest real-time streaming clinical updates without full index recalculation. Furthermore, multi-modal medical imaging was excluded from graph nodes.

5. Conclusion

While Graph RAG establishes superior factuality bounds for complex medical queries, the engineering overhead and latency overhead necessitate hybrid routing mechanisms.